Smart Lab - status report

Generated 2026-11-03 04:41 on the portal laptop. Live values were read at that moment (read-only); nothing was changed to produce this report.

1. Portal state

Item	State
Backend health (/api/health)	HTTP 000
backup	Up 13 hours
db	Up 13 hours (healthy)
frontend	Up 2 hours
n8n	Up 13 hours
AI assistant model (ollama ps)	qwen2.5:1.5b 65ec06548149 1.1 GB 100% CPU 2048 llamacpp 23 hours from now / nomic-embed-text:latest 0a109f422b47 389 MB 100% CPU 2048 llamacpp About an hour from now
Assistant GPU gate	{"error": "missing or wrong X-KB-Key"}

2. Lab PCs

Device	Name	IP	State	Last seen	Mode	Role	Lab
PC_01	PC-01		offline	2026-10-21 05:16	NORMAL		LAB_01
PC_ADMIN_LAPTOP	ADMIN-LAPTOP		online	2026-11-03 03:51			LAB_01

3. Other devices (ESP32 nodes, cameras, robots)

Device	Name	Type	IP	State	Last seen	Lab
CAM_LAB01	Entry Camera (ESP32-CAM)	CAMERA	192.168.1.150	offline	never	LAB_01
FACESRV_01	Face Recognition Server	FACE_SERVER	192.168.1.200	offline	never	LAB_01
MASTER_LAB01	Door Controller (Master ESP32)	MASTER_CONTROLLER	192.168.1.151	offline	2026-11-02 03:28	LAB_01
FANUC_01	FANUC ER-4iA (physical)	ROBOT		online	2026-11-03 03:51	LAB_01
FANUC_SIM	FANUC (simulation)	ROBOT		offline	2026-10-08 17:38	LAB_01
UR5E_01	UR5e (physical)	ROBOT		offline	2026-11-03 03:45	LAB_01
UR5E_SIM	UR5e (simulation)	ROBOT		offline	2026-10-08 19:49	LAB_01
SECCAM_LAB01	Security camera	SECURITY_CAMERA		offline	2026-11-03 03:45	LAB_01
SECCAM_LAB02	Security camera	SECURITY_CAMERA		offline	2026-11-03 03:45	LAB_01
BIN_NODE_1	BIN_NODE_1	SENSOR_NODE	192.168.1.153	offline	2026-11-01 23:37	LAB_01
ENV_LAB01	Environment node	SENSOR_NODE	192.168.1.152	offline	2026-10-30 20:02	LAB_01
REUSE_BIN_1	REUSE_BIN_1	SENSOR_NODE	192.168.1.154	offline	2026-10-30 20:02	LAB_01

Offline here is expected at home: the robots, lab cameras and ESP32 boards are not connected tonight.

4. Services on this laptop (scripts\status.ps1)

```
Smart Lab services  04:42:22   portal health: no answer
  face_server       UP
  security_camera   UP     [ WARN:0@2685.266] global cap_ffmpeg_impl.hpp:453 _opencv_ffmpeg_inter...
  security_camera2  UP     [ WARN:0@2702.984] global cap_ffmpeg_impl.hpp:453 _opencv_ffmpeg_inter...
  ur5e_monitor      UP     connect: A connection attempt failed because the connected party did n...
  fanuc_monitor     UP     [robot] read failed: ConnectTimeout: HTTPConnectionPool(host='192.168....
  ollama            UP     (no log yet)
  knowledge         UP     [kb-service] 127.0.0.1 "GET /status HTTP/1.1" 200 -
  voice             UP     ===== 2026-11-03 02:36:30 start: py -3.12 C:\Users\dell\OneDrive\Deskt...
  opcua             UP     2026-11-03 03:12:03,711 WARNING feed error: ReadTimeout
  pc_agent          UP     ===== 2026-11-03 02:36:32 start: call C:\Users\dell\OneDrive\Desktop\B...
  lan_presence      UP     ===== 2026-11-03 02:36:34 start: call C:\Users\dell\OneDrive\Desktop\B...
  mdns              UP     Announcing https://smartlab.local -> 192.168.1.200  (close this window...
```

5. Network + lab pre-flight (scripts\lab_preflight.ps1)

```
Smart Lab pre-flight  2026-11-03 04:44:21  on ALI
  Network   Smart Lab address 192.168.1.200           PASS   on 'Ethernet' (Preferred)
  Network   No gateway on 'Ethernet'                  PASS   internet goes out the other adapter
  Network   Robots reached over the cable             WARN   192.168.1.102 leaves via 'Wi-Fi' (want 'Ethernet'; both nets 192.168.1.x at
  Network   Internet adapter                          PASS   Wi-Fi 192.168.1.8 via 192.168.1.1 (portal works without it)
  Firewall  'Smart Lab ports' inbound rule            FAIL   missing - run firmware\face_server\allow_firewall.bat once as administrator
  Firewall  Profile of 'Wi-Fi'                        INFO   Public (the rule above uses profile=any, so Public is fine)
  Firewall  Profile of 'Ethernet'                     INFO   Public (the rule above uses profile=any, so Public is fine)
  Portal    Backend health :8000                      FAIL   HTTP 0
  Portal    Web :80 (redirects to https)              FAIL   HTTP
  Portal    Docker services                           PASS   backup=running db=running frontend=running n8n=running
  Services  HTTPS (phones, PCs) :443                  PASS   listening
  Services  Face server (ESP32-CAM) :5000             WARN   not listening - start launch.bat (its window shows why)
  Services  Security camera 1 :5100                   WARN   not listening - start launch.bat (its window shows why)
  Services  Security camera 2 :5101                   PASS   listening
  Services  Ollama (assistant) :11434                 WARN   not listening - start launch.bat (its window shows why)
  Services  Knowledge service :5199                   WARN   not listening - start launch.bat (its window shows why)
  Devices   Smart Lab router (Huawei) 192.168.1.1     PASS   ping ok, tcp 80 open
  Devices   FANUC ER-4iA controller 192.168.1.100     PASS   ping ok, tcp 80 closed
  Devices   UR5e controller (RTDE) 192.168.1.102      FAIL   ping no reply, tcp 30004 closed
  Devices   ESP32-CAM (door face) 192.168.1.150       FAIL   ping no reply
  Devices   Master ESP32 (door) 192.168.1.151         FAIL   ping no reply
  Devices   Environment node 192.168.1.152            FAIL   ping no reply
  Devices   Security camera 1 192.168.1.155           FAIL   ping no reply, tcp 554 closed
  Devices   Security camera 2 192.168.1.156           FAIL   ping no reply, tcp 554 closed
  9 FAIL, 5 WARN
  saved C:\Users\dell\OneDrive\Desktop\Bsc\codes\smart-lab-portal\logs\preflight-20261103-044421.txt
```

6. Updates in this session

Area	Change	Verified
Launcher	launch.bat opens a visible window for every service again (your request). Hidden version kept in backups/launch.bat.hidden-2026-11-03.bak. New scripts/status.ps1 shows UP/DOWN per service + portal health.	14 window starts; status.ps1 run
Launcher	Shows 192.168.1.200 as 'this laptop' (was the Wi-Fi/hotspot address, false warning on screen).	tested -> 192.168.1.200
Launcher	scripts/bg.ps1 (hidden mode helper): quoted paths with spaces + 'pause' no longer hang (found by Gemini, approved).	scratch test
Detector	Two classes on the same object keep only the more confident (master was also called 'router 81%'). A small object in front of a big one is kept.	16/16 detector tests
Detector	Switch and router need 70% (was 50%) - removes the false 'switch 67%' on the floor; real ones scored 83-92%.	applied to detectors.json
Scan page	Hint shows the device name, not 'SLAB:D:MASTER_LAB01'. The phone's generic 'laptop' box is hidden over a lab device.	tsc 0 errors
Scan page	Phone heat/freeze: QR checks 8/s -> 4/s, people frames at most every 0.8 s (were back-to-back).	tsc 0 errors
Assistant	'Tell me more' slow: the model ran 100% on CPU. Turned on your camera-first auto mode (GPU while cameras idle, CPU the moment a camera analyses). Backups in backups/gpu-auto-20261103/.	applies after restart
Network	PC agent + LAN presence prefer the 192.168.1.x address on dual-network PCs (Gemini FIX-1).	new test, 6 + 14 pass
Network	NEW tools/relay/relay.py for the lab engineer PC: GIU PCs (10.40.20.x) reach the portal through it - without it their agents cannot connect at all.	3 tests + live: API 200, HTTPS 200
Network	Firewall script now opens 80, 443, 5000, 5100, 8000, 8080, 8443 (not run - needs admin).	file checked
Network	NEW scripts/lab_preflight.ps1: read-only check of addresses, routes, firewall, portal, services, robots/cameras/ESP32.	run tonight
Docs	NEW docs/LAB_DAY_2026-11-04.md: step-by-step lab day, who does what, risks, Gemini review table.	written
Training	Switch video: 51 frames in tools/vision_train/raw/switch_video_2026-11-03 (needs other backgrounds too).	contact sheet checked
Collaboration	Automatic Gemini loop tested end-to-end (APPROVED), then turned OFF at your request; Antigravity remote-control daemon + autostart disabled (backups kept).	processes verified

7. Tests run

Suite	Result
Backend detector tests	16 passed (after fixing my own over-eager overlap rule that one old test caught)
tools/security_vision	292 passed
tools/knowledge	49 passed
tools/robot_monitor	42 passed
tools/pc_agent	14 passed
tools/lan_presence	6 passed
tools/relay	3 passed + live relay test
Frontend type-check (tsc)	0 errors
Full backend suite (730 tests)	passed 730 / 2 skipped on 2026-11-02; NOT re-run after tonight's small detector change (only the detector file changed - its 16 tests pass)

8. Gemini network audit - checked against the code

Gemini finding	Verdict	Action
FIX-1 dual-NIC PC lists GIU IP first	correct	fixed in pc_agent, win7 copy, lan_presence
FIX-2 firewall lacks 443	correct	added 443/8080/8443; 4840 and 5101 not added (localhost only)

Gemini finding	Verdict	Action
FIX-3 launcher shows wrong IP	correct	fixed, tested
FIX-4 reusable bin LAB_ID BENCH	correct	reported only - firmware is owner-protected
FIX-5 rebuild + full launch	correct	in progress (see Issues)
ROUTER-1 DHCP pool .10-.90	correct (177 log lines with .101)	owner sets on the router
Never run win7_usb install.bat on GIU PCs	correct	in the lab-day steps
Postgres 0.0.0.0:5432 / Redis / Caddy	WRONG	Postgres is 127.0.0.1:5433; no Redis or Caddy exist
Other lab PCs stay on GIU only, it all works	INCOMPLETE	they have no route to the portal - relay added

9. Needs a person tomorrow (not code)

Item	Who
ISSUE - laptop memory: 1.2 GB free of 15.8 GB, 6.6 GB page file in use, ~230 pages/s swapping (measured 03:58). Docker slows ~20x: backend import 18 s instead of ~1 s, DB dump 6 min, restart up to 10 min, image build >20 min. Closed ChatGPT, Notion, Notion Calendar, Copilot, Teams, MATLAB helpers, Gemini app -> 2.4 GB free. Next: close Chrome tabs / extra VS Code windows; optional: lower Docker's 8 GB memory cap (needs a Docker restart - your OK).	owner / Claude
ISSUE - on battery the laptop throttles (seen tonight). Always on the charger at the lab.	owner
ISSUE - launcher 'up' sometimes hangs swapping the backend (new container stuck at 'Created', old one stopped; seen 03:00 and 03:50). Fix tonight: build first, then start; recreate under memory pressure is the trigger.	Claude (in progress)
Deploy of tonight's changes: image build running (slow - memory). Portal will be live after 'docker compose up'; then launch.bat attaches.	Claude (in progress)
Run firmware/face_server/allow_firewall.bat once as administrator	owner
Router DHCP pool -> 192.168.1.10-.90 before the robots are powered	owner
Engineer PC: USB-Ethernet 192.168.1.210, no gateway; firewall inbound 8000/443/80 for the relay	lab engineer / GIU IT
Videos: UR5e, FANUC, both pendants, switch on other backgrounds, 30 s clips from both lab cameras	owner
Your earlier requests still open: responsive audit needs E2E passwords; commit of all work needs your OK (nothing committed or pushed)	owner

Full lab-day steps: docs/LAB_DAY_2026-11-04.md. Nothing in this session was committed or pushed.